# VARUN BAJPAI

Full Stack Developer | MERN | AI/RAG | CSE Undergraduate
Meerut, UP | varunbajpai16@gmail.com | Available for Internship | LinkedIn | GitHub | LeetCode

## EDUCATION

Bachelor of Technology (B.Tech) — Computer Science & Engineering
Meerut Institute of Engineering and Technology, Meerut | 3rd Year | CGPA: 7.4 / 10

## TECHNICAL SKILLS

Languages: C++, JavaScript
Frontend: React.js, Redux Toolkit, Tailwind CSS, HTML5, CSS3, Responsive Design
Backend: Node.js, Express.js, REST APIs, JWT Authentication, Express Middleware
Databases: MongoDB, PostgreSQL, Vector Databases
AI: LangChain, LangGraph, RAG, Agentic AI
Tools & Platforms: Git, GitHub, Postman, Vercel, Render, Cloudinary
Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems

## PROJECTS

**TutorMate — MERN Stack | Live Demo | GitHub**
• Developed and deployed a full-stack tutor discovery platform serving 100+ users with Google OAuth and role-based access for Teachers, Students, and Parents.
• Implemented 2dsphere geospatial queries in MongoDB for proximity-aware teacher discovery; designed 10+ REST APIs with Express.js and Mongoose.
• Built a responsive React.js/Tailwind CSS interface and deployed frontend on Vercel and backend on Render.

**YouTube Clone — MERN Stack | Live Demo | GitHub**
• Built a full-stack video streaming platform supporting upload, playback, likes, comments, and subscriptions for 50+ test users.
• Designed scalable MongoDB schemas and 15+ RESTful APIs; implemented JWT authentication with refresh tokens and Cloudinary media delivery.

**NexusAI — AI / RAG Project | Live Demo | GitHub**
• Built and deployed a RAG application using LangChain, LangGraph, and vector databases to retrieve relevant context and generate contextual responses.

**EcoTrust — Environmental Intelligence Platform | Live Demo**
• Built an environmental monitoring platform that ingests simulated CEMS sensor data to track industrial emissions, compliance, and factory-level alerts.
• Implemented data validation and trust scoring using completeness, consistency, sensor reliability, temporal stability, and validation history.
• Developed a role-based React dashboard with geospatial factory monitoring and AI/agentic workflows using LangChain/LangGraph, backed by Node.js, Express, and MongoDB.

## LEADERSHIP & ACTIVITIES

• Smart India Hackathon (SIH) — Team Lead: Led a 6-member cross-functional team to design and prototype a government problem-statement solution under 36-hour hackathon constraints.
• Coordinated task distribution, sprint planning, technical execution, and final presentation to a panel of industry judges.
• Open Source & Self-Initiated Projects: Independently architected, built, and deployed MERN applications publicly using Vercel and Render.

## ACHIEVEMENTS

• Solved 200+ DSA problems on LeetCode across arrays, trees, graphs, and dynamic programming.
• Built and deployed full-stack MERN applications independently from architecture through production deployment.

## PROFILES & PROJECT LINKS

LinkedIn: https://www.linkedin.com/in/varun-bajpai16/
GitHub: https://github.com/varunbajpai16-sudo
LeetCode: https://leetcode.com/u/VarunBajpai17/
TutorMate / MentorOwl: https://mentor-owl.vercel.app/
YouTube Clone: https://yt-project-six.vercel.app/
NexusAI: https://nexus-ai-kappa-flax.vercel.app/
EcoTrust: https://eco-trust-peach.vercel.app/